Notes: Garicano and Rossi-Hansberg (QJE, 2006)

Organization and Inequality in a Knowledge Economy

Contents

1 Big picture

- **Question.** How does the organization of work affect the distribution of knowledge, wages, firm size, layers of management, and wage inequality in a knowledge economy?
- **Core idea.** Production requires solving problems. Agents differ in cognitive ability. More able agents can learn more cheaply, but their time is scarce. Hierarchies economize on expert time by letting less able agents handle routine problems and pass exceptional problems upward.
- **Main organizational object.** A **knowledge hierarchy**: workers produce and solve routine problems; managers solve more difficult exceptions; higher layers solve still rarer and harder problems.
- **Main theoretical contribution.** The paper embeds the single-hierarchy logic of Garicano-style knowledge hierarchies into a general equilibrium assignment model with heterogeneous ability, endogenous knowledge acquisition, endogenous occupations, endogenous matching, and endogenous wages.
- **Why hierarchy matters for inequality.** Organization lets high-skill agents leverage their knowledge over many problems generated by lower layers. It also lets low-skill workers substitute asking for learning. Therefore, relative to autarky, organization raises the marginal value of top ability and lowers the marginal value of bottom ability.
- **Equilibrium sorting result.** The model produces **positive sorting**: higher-ability managers are matched with higher-ability subordinates. Better subordinates shield better managers from easier questions, letting expert time be used only on harder problems.
- **Equilibrium stratification result.** Ability is segmented by occupational rank. The least able agents become production workers, the next agents become low-level managers, higher agents become higher-level managers, and the most able agents become entrepreneurs at the top of hierarchies.
- **Wage result.** The equilibrium wage function is increasing and convex in ability. Wage convexity follows because more able agents acquire more knowledge and knowledge is more valuable when it can be leveraged over a hierarchy.
- **Two distinct types of information technology.** The paper distinguishes (i) reductions in the cost of **communicating** knowledge, denoted by lower h, and (ii) reductions in the cost of **acquiring** knowledge, denoted by lower t.
- **Comparative static 1: lower communication cost.** Lower h makes asking cheaper. The economy uses organization more intensively, has more layers and larger spans, workers learn less, top managers learn more, routine-worker inequality falls, and top inequality rises.
- **Comparative static 2: lower learning cost.** Lower t makes acquiring knowledge cheaper. Agents at all layers learn more, production becomes more knowledge-intensive, decentralization rises, and within-layer as well as overall wage inequality rises.
- **Empirical motivation.** The paper uses U.S. wage and organization patterns as suggestive evidence. The 1980s and early 1990s look like an era of cheaper information processing; the late 1990s look more like an era of cheaper communication.

- **Takeaway.** Wage inequality cannot be understood only from worker skills or technologies in isolation. It also depends on how knowledge is organized, who learns, who asks, who answers, and how expert time is leveraged.

2 Data and measurement (key objects)

2.1 Why this is mainly a theory paper

- The paper is not a standard empirical paper with a treatment, an instrument, or a causal regression design.
- The central contribution is a formal equilibrium model of knowledge acquisition, communication, hierarchy, assignment, and wages.
- The empirical section is mainly a consistency check against broad U.S. facts about wage inequality, CEO pay, firm size, and organizational flattening.

Interpretation. The paper should be read as a theory of how information technology affects both organization and inequality. The empirical evidence is suggestive and illustrative rather than a decisive causal test.

2.2 Agents, ability, and problem solving

Agents are heterogeneous in cognitive ability:

$$\alpha \sim \phi(\alpha), \qquad \alpha \in [0, 1]. \tag{1}$$

A higher value of α means lower cost of acquiring knowledge. Agents have one unit of time. Time can be used either in production or in helping others solve problems.

Problems are ordered by how common they are. A problem difficulty level is denoted by Z. The density of problems is $f(Z)$ and the distribution is $F(Z)$, with more common problems ordered first. Knowledge is cumulative: an agent who can solve harder problems also knows all easier problems.

The paper works with the fraction of problems an agent can solve:

$$q = F(\tilde{z}), \qquad \tilde{z} = z(q) = F^{-1}(q). \tag{2}$$

Here $z(q)$ is the amount of knowledge required to solve fraction q of all problems. Because rarer problems are harder to reach, $z'(q) > 0$ and $z''(q) > 0$.

Interpretation. The variable q is a convenient index of problem-solving scope. A worker with $q = 0.3$ can solve the most common 30 percent of problems.

2.3 Cost of acquiring knowledge

The cost per unit of knowledge is:

$$c(\alpha; t) = t - \alpha. \tag{3}$$

Therefore, the cost of learning enough to solve fraction q of problems is:

$$c(\alpha; t)z(q) = (t - \alpha)z(q). \tag{4}$$

The parameter t measures the general cost of accessing or acquiring knowledge. A reduction in t is an improvement in information-processing or knowledge-acquisition technology.

Interpretation. A lower t makes it cheaper for everyone to learn. A higher α makes a particular agent better at learning.

2.4 Communication cost and hierarchy

A production worker draws one problem per unit of production time. If she can solve it, output is produced. If she cannot solve it, she can ask a manager above her.

Each communicated problem costs the manager $h < 1$ units of time:

$$h = \text{time cost of communicating and diagnosing one problem.} \tag{5}$$

A reduction in h is an improvement in communication technology, such as cheaper e-mail, mobile communication, or lower coordination costs.

Interpretation. Lower h means experts can handle more questions per unit of time. This raises their span of control and increases the leverage of expert knowledge.

2.5 Layers, workers, managers, and entrepreneurs

A hierarchy has layers $l = 0, 1, \ldots, L$.

- Layer 0 agents are production workers. They produce and draw problems.
- Layers $1, \ldots, L-1$ are managers. They solve problems passed up from below.
- Layer L is the top layer. The top manager is called the entrepreneur.

If layer l agents have knowledge q_l, then layer $l+1$ is asked about problems that layer l cannot solve. The hierarchy is pyramidal because only a fraction of problems moves upward.

The time constraints are:

$$hn_0(1 - q_{L-1}) = n_L \equiv 1, \tag{6}$$
$$hn_0(1 - q_{L-2}) = n_{L-1}, \tag{7}$$
$$\vdots \tag{8}$$
$$hn_0(1 - q_0) = n_1. \tag{9}$$

Equivalently,

$$n_l = hn_0(1 - q_{l-1}), \qquad l = 1, \ldots, L. \tag{10}$$

Total expected output of a hierarchy is:

$$y = q_L n_0. \tag{11}$$

Interpretation. More knowledgeable lower layers ask fewer questions, so higher layers can supervise more workers. More knowledgeable top layers raise the probability that a problem is ultimately solved.

2.6 U.S. empirical facts used for motivation

The paper uses several broad U.S. patterns:

- rising within-occupation wage inequality in the 1980s and early 1990s;
- a later pattern in which upper-tail inequality keeps rising while lower-tail inequality slows or declines;
- a sharp rise in the CEO-worker pay ratio;
- evidence from prior work that organizations became flatter and more decentralized in the 1980s and early 1990s;
- a negative correlation between mean log firm size and the standard deviation of log hourly wages in the authors' time-series plot.

Interpretation. These facts motivate the distinction between cheaper knowledge acquisition and cheaper communication. They are not used as formal identification evidence.

3 Models and empirical specifications (all equations + interpretation)

3.1 The firm problem

A hierarchy with L layers chooses abilities α_l, knowledge levels q_l, and layer sizes n_l. Profits are output minus wages and learning costs:

$$\Pi^{(L)} = \max_{\{q_l, n_l, \alpha_l\}_{l=0}^{L}} q_L n_0 - \sum_{l=0}^{L} n_l \left[c(\alpha_l; t) z(q_l) + w(\alpha_l) \right], \tag{12}$$

subject to the time constraints:

$$hn_0(1 - q_{l-1}) = n_l, \qquad l = 1, \ldots, L, \tag{13}$$

with $n_L = 1$ for normalization.

The first-order condition for the ability of employees implies:

$$w'(\alpha) = -c_\alpha(\alpha; t) z(q). \tag{14}$$

Because $c(\alpha; t) = t - \alpha$, we have $c_\alpha(\alpha; t) = -1$. Hence,

$$w'(\alpha) = z(q(\alpha)). \tag{15}$$

Interpretation. The marginal return to ability equals the amount of knowledge the job requires. If a technological change makes agents acquire more knowledge, the wage schedule becomes steeper.

3.2 The agent problem and incentive compatibility

An agent of ability α can apply for a job designed for ability α'. A job α' pays $w(\alpha')$ and requires knowledge $q(\alpha')$. The agent's problem is:

$$U(\alpha) = \max_{\alpha'} w(\alpha') + \left[c(\alpha'; t) z(q(\alpha')) - c(\alpha; t) z(q(\alpha')) \right]. \tag{16}$$

The term in brackets adjusts the compensation for the difference between the learning cost of the job-design type α' and the actual learning cost of the worker α.

In equilibrium, agents choose the job designed for their own ability. The wage slope condition from the firm problem supports this incentive compatibility.

Interpretation. Wages not only allocate agents to positions. They also give agents the right incentives to acquire the knowledge required by those positions.

3.3 Equilibrium objects

A competitive equilibrium contains:

- a set of operating hierarchy sizes $\bar{L}$;
- occupational sets A_l^M and A_l^E, where M denotes managers and workers, and E denotes entrepreneurs;
- a wage function $w(\alpha)$;
- an assignment function $a(\alpha)$ that maps a subordinate to a supervisor;
- a knowledge function $q(\alpha)$;
- a span-of-control function $n(\alpha)$ giving direct subordinates.

Market clearing requires that the supply of employees in any ability interval equal the demand for employees by the managers and entrepreneurs assigned to that interval.

Interpretation. The equilibrium is not just a wage schedule. It jointly determines who learns what, who works for whom, how many people report to each manager, how many layers exist, and how income is distributed.

3.4 Positive sorting and the assignment equation

The paper proves that any equilibrium has positive sorting. Better managers are matched with better subordinates.

Let $a(\alpha)$ denote the supervisor assigned to an agent of ability α. The assignment function satisfies an ordinary differential equation. In simplified form:

$$\frac{\partial a(\alpha)}{\partial \alpha} = \begin{cases} \dfrac{1 - q(\alpha)}{1 - q(a^{-1}(\alpha))} \dfrac{\phi(\alpha)}{\phi(a(\alpha))}, & \alpha \in A_M \setminus A_0^M, \\ h(1 - q(\alpha)) \dfrac{\phi(\alpha)}{\phi(a(\alpha))}, & \alpha \in A_0^M. \end{cases} \tag{17}$$

Interpretation. The slope of the assignment function depends on the span of control implied by knowledge choices and on the density of agents at different ability levels. Better subordinates lower the flow of easy problems to managers, so they are especially valuable to better managers.

3.5 Wage convexity

The wage function is continuous, increasing, and convex.

Using the wage slope condition:

$$w'(\alpha) = z(q(\alpha)) > 0. \tag{18}$$

Where differentiable,

$$w''(\alpha) = z'(q(\alpha))q'(\alpha). \tag{19}$$

Since $z'(q) > 0$ and equilibrium knowledge $q(\alpha)$ is increasing in ability, the wage function is convex within occupations. The paper also shows global convexity across occupational thresholds.

Interpretation. The model generates a convex earnings schedule because higher ability is rewarded twice: high ability lowers learning costs, and organization lets the resulting knowledge be used on more problems.

3.6 Occupational stratification

An equilibrium with maximum number of layers L is characterized by thresholds:

$$\{\alpha^*_{ll}, \alpha^*_{l,l+1}\}_{l=0}^{L}. \tag{20}$$

The least able agents are workers:

$$[0, \alpha^*_{00}]. \tag{21}$$

Intermediate ability agents are managers:

$$[\alpha^*_{00}, \alpha^*_{L-1,L-1}]. \tag{22}$$

The highest ability agents are entrepreneurs:

$$[\alpha^*_{L-1,L-1}, 1]. \tag{23}$$

The equilibrium can sustain only hierarchies with two adjacent numbers of layers:

$$\bar{L} = \{L-1, L\}. \tag{24}$$

Interpretation. The model does not predict arbitrary mixing of workers and managers across abilities. It predicts stratification by rank: routine production at the bottom and increasingly exceptional problem solving at the top.

3.7 Organization versus autarky

Without organization, each agent solves only her own problems. With organization, high-skill agents can help many lower-skill agents.

Relative to autarky:

- production workers acquire less knowledge because they can ask for help;
- entrepreneurs acquire more knowledge because they can leverage their knowledge over a larger set of problems;
- the marginal value of bottom skill falls;
- the marginal value of top skill rises.

Interpretation. Organization is inequality-amplifying at the extremes. It lowers the need for routine workers to learn but raises the reward to experts who can solve rare problems.

3.8 Knowledge transactions and transfers

The paper shows that the hierarchy can be interpreted in several equivalent ways:

- as an integrated firm with internal wage transfers;
- as a referral market in which agents sell unsolved problems upward;
- as a consulting market in which agents pay more knowledgeable experts for help.

A central recursive object is $p(q_l; w)$, the transfer or fee associated with passing problems upward. For an intermediate manager, earnings can be written as:

$$w(\alpha_l) = \frac{q_l - q_{l-1} + p(q_{l-1}; \cdot) - p(q_l; \cdot)}{h(1 - q_{l-1})} - c(\alpha_l; t)z(q_l). \tag{25}$$

Interpretation. A manager is paid for the problems she solves, pays to pass problems she cannot solve, and receives or pays transfers that coordinate the hierarchy. The model's logic is about knowledge transactions, not only formal firm boundaries.

3.9 Numerical exercise

The paper studies numerical examples with:

$$f(z) = \lambda e^{-\lambda z}, \qquad \lambda = 2, \tag{26}$$

and a uniform ability distribution:

$$\alpha \sim U[0, 1]. \tag{27}$$

The examples vary two parameters:

- h, the communication cost;
- t, the cost of acquiring knowledge.

The numerical graphs show the equilibrium wage and knowledge functions for combinations of high and low h and high and low t.

Interpretation. Because the full equilibrium has jointly determined wages, knowledge, matching, spans, and layers, the paper uses numerical exercises to illustrate comparative statics that are difficult to prove analytically in full generality.

4 Identification and credibility (what makes the argument convincing)

4.1 No instrument and no causal regression design

- The paper does not use an instrumental variable, difference-in-differences design, regression discontinuity, or randomized experiment.
- Identification is not the right language for the paper's main contribution.
- The main credibility comes from a coherent equilibrium model, formal propositions, and qualitative consistency with broad empirical patterns.

Interpretation. The paper's standard is theoretical discipline rather than causal estimation.

4.2 Formal results that discipline the model

The model proves several nontrivial equilibrium properties:

- equilibrium involves positive sorting;

- equilibrium wages are increasing and convex;
- equilibrium knowledge is increasing in ability;
- organization raises top knowledge and lowers worker knowledge relative to autarky;
- occupations are stratified by ability;
- equilibrium exists and is unique;
- the equilibrium allocation is Pareto optimal.

Interpretation. These results prevent the model from being only a verbal story. The comparative statics are tied to a precise equilibrium structure.

4.3 What distinguishes communication technology from knowledge-acquisition technology

The paper's strongest conceptual identification is the separation of two technologies:

Technology shock	Immediate economic meaning	Organizational implication
Lower h	Asking and communicating become cheaper	More hierarchy, more centralization, larger spans, more layers
Lower t	Learning and accessing knowledge become cheaper	More learning at lower levels, more decentralization, more self-employment

Interpretation. The same broad phrase "information technology" can mean two different things with opposite effects on organization and parts of the wage distribution.

4.4 Limits of the evidence

- The U.S. evidence is based on broad aggregate patterns and prior empirical studies.
- The paper does not isolate exogenous changes in h or t.
- Firm size is used as a proxy for hierarchy size, but the model determines hierarchy boundaries rather than legal firm boundaries.
- The model abstracts from cross-country teams, multiple labor markets, and other functions of hierarchy such as monitoring, incentives, and authority.

Interpretation. The evidence is suggestive. The main contribution is a framework that organizes facts and generates testable implications.

4.5 Relation to the broader literature

- Compared with standard skill-biased technical change, the model emphasizes organization and hierarchy, not only skill demand.

- Compared with superstar models, top agents become high earners because communication and organization let them leverage their knowledge over more production problems.
- Compared with one-to-one matching models, the paper studies multi-sided, one-to-many hierarchical matching.
- Compared with single-team hierarchy models, the paper puts hierarchy choice into a labor-market equilibrium with endogenous wages and assignment.

Interpretation. The paper connects inequality, assignment, organization, and information technology in one equilibrium framework.

5 Tables and figures

5.1 Figure I: Assignments in equilibrium

Read.

- Ability is arranged on a line from low to high.
- Arrows indicate matching from subordinates to supervisors.
- The assignment function is increasing: low-ability workers are matched with low-ability managers; higher-ability workers are matched with higher-ability managers or entrepreneurs.
- Occupational thresholds divide workers, managers, and entrepreneurs.

Economic message. The hierarchy is not random. It is positively sorted. High-skill supervisors want high-skill subordinates because such subordinates shield them from routine problems.

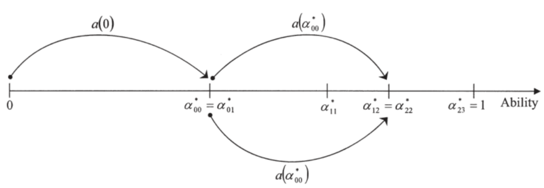

Figure I
Assignments in Equilibrium

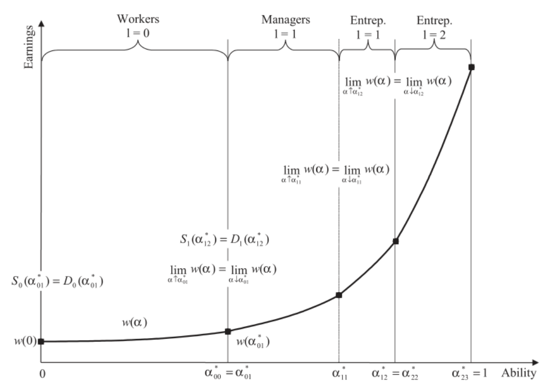

Figure II
Construction of an Equilibrium Allocation

5.2 Figure II: Construction of an equilibrium allocation

Read.

- The wage function is increasing and convex.

- Boundaries between workers, managers, and entrepreneurs are marked by thresholds.
- At threshold points, wages are continuous but the slope can change because knowledge requirements jump.

Economic message. Equilibrium wages must make agents willing to choose the occupation and knowledge level designed for their ability. Convexity reflects the increasing leverage of knowledge at higher ability levels.

5.3 Figure III: Numerical examples

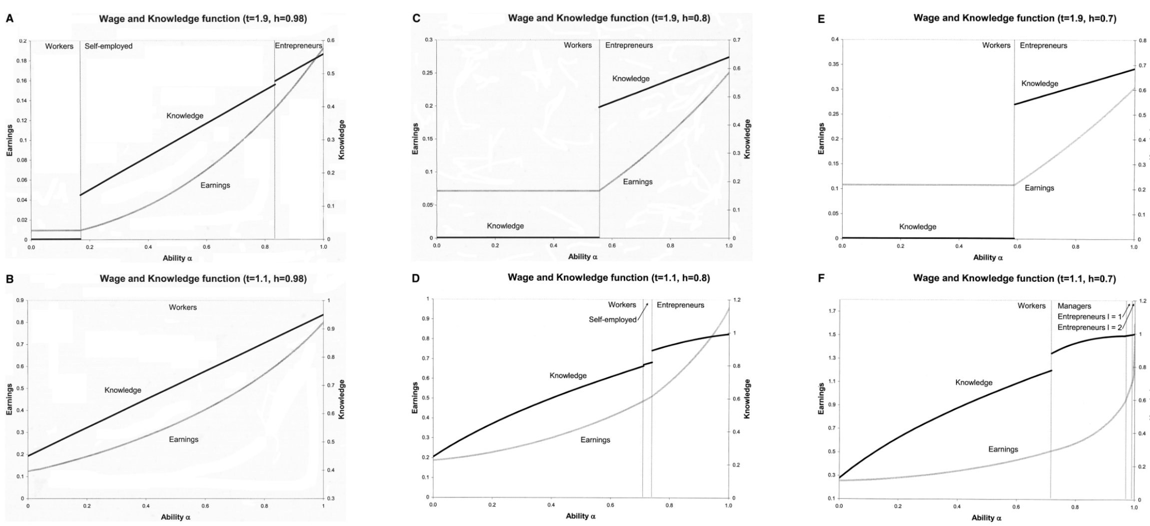

FIGURE III
Numerical Examples

FIGURE III
Numerical Examples (continued)

FIGURE III
Numerical Examples (continued)

Read.

- Each panel plots the equilibrium earnings and knowledge functions across ability.
- Moving from high h to low h makes communication cheaper.
- Moving from high t to low t makes learning cheaper.
- Lower communication costs increase reliance on higher layers and can create more pronounced top inequality.
- Lower learning costs raise knowledge at all levels and steepen the wage schedule.

Economic message. The numerical examples show why communication and information-processing improvements have different effects. Cheaper communication centralizes problem solving; cheaper learning decentralizes it.

5.4 Figure IV: Firm size and wage inequality

Read.

- The figure plots mean log firm size and a scaled measure of log hourly wage dispersion.

- The two series move strongly in opposite directions over the sample shown.
- The figure reports a correlation close to -0.95.

Economic message. The paper interprets the negative comovement as suggestive evidence that technological changes affecting knowledge and communication may jointly move organization and wage inequality.

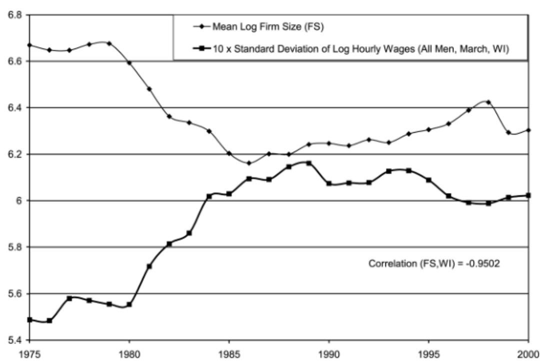

FIGURE IV
Firm Size and Wage Inequality
Source: Mean Log Firm Size: Compustat, SD of Log Hourly Wages: March CPS using the methodology in Card and DiNardo [2002].

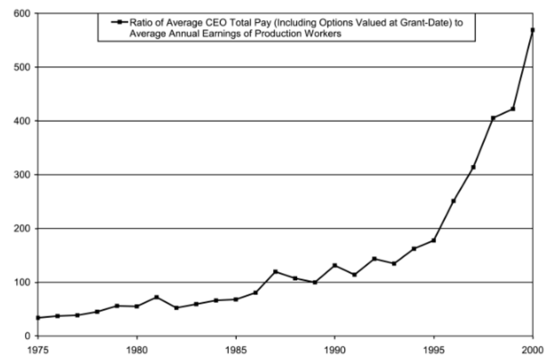

FIGURE V
Worker-Manager Wage Inequality
Source: CEO sample is based on all CEOs included in the S&P 500, using data from Forbes and ExecuComp. CEO total pay includes cash pay, restricted stock, payouts from long-term pay programs, and the value of stock options granted using Execu-

5.5 Figure V: Worker-manager wage inequality

Read.

- The CEO-worker pay ratio rises sharply, especially in the late 1990s.
- This is a stark upper-tail inequality fact.

Economic message. The model can rationalize rising top inequality through the increased leverage of high-level problem solvers when communication costs fall and organizations become larger or more layered.

6 Comparative statics: what changes when technology changes

6.1 Lower communication cost h

A reduction in h means that managers can receive and diagnose more problems per unit of time. The cost of asking falls relative to the cost of learning.

Organization.

- Hierarchies become more attractive.
- The share of self-employed agents falls.
- Spans of control rise.

- The number of layers can increase.
- More problems are solved at higher levels.

Knowledge.

- Production workers learn less because asking is cheaper.
- Entrepreneurs learn more because their knowledge can be leveraged over more problems.
- The economy becomes more centralized in problem solving.

Inequality.

- Within-worker inequality falls because workers acquire less knowledge and the wage slope is lower at the bottom.
- Top inequality rises because managers and entrepreneurs acquire more knowledge and leverage it over larger teams.
- Manager-worker inequality rises.
- Overall inequality can initially fall and later rise depending on whether the lower-tail equalization or upper-tail amplification dominates.

Economic intuition. Cheap communication creates a superstar-like effect for high-level problem solvers. Their expertise becomes useful over a larger mass of production problems.

6.2 Lower cost of acquiring knowledge t

A reduction in t means that learning becomes cheaper for everyone.

Organization.

- Self-employment becomes more attractive because agents can solve more problems themselves.
- Production becomes more knowledge-intensive.
- If communication costs are high, the number of layers may fall.
- Spans of control can rise because subordinates ask fewer questions.

Knowledge.

- Workers acquire more knowledge.
- Managers acquire more knowledge.
- More problems are solved closer to the production floor.
- The economy becomes more decentralized.

Inequality.

- Within-worker inequality rises.
- Within-manager inequality rises.

- Overall wage inequality rises because ability differences become more valuable when everyone learns more.

Economic intuition. Cheap learning increases the knowledge content of jobs. Since high-ability agents have comparative advantage in learning, lower t steepens wages.

6.3 Summary table

	Lower communication cost h	**Lower knowledge-acquisition cost t**
Direct effect	Asking becomes cheaper	Learning becomes cheaper
Organization	More hierarchy, more centralization	More self-sufficiency and decentralization
Workers' knowledge	Falls	Rises
Top managers' knowledge	Rises	Rises
Layers	More likely to rise	More likely to fall when h is high
Span of control	Rises	Can rise because workers ask less
Worker inequality	Falls	Rises
Top inequality	Rises	Rises
Overall message	Experts are leveraged more	Jobs become more knowledge-intensive

7 Short summary

- The paper studies a knowledge economy in which heterogeneous agents decide how much knowledge to acquire and whether to work as workers, managers, or entrepreneurs.
- Knowledge is cumulative, problems differ in frequency, and communication allows agents to form hierarchies.
- Hierarchies economize on expert time: workers solve routine problems and pass exceptional problems upward.
- Equilibrium features positive sorting, occupational stratification, increasing knowledge in ability, and a convex wage function.
- Organization reduces workers' incentives to learn but increases top managers' incentives to learn.
- Lower communication costs centralize problem solving and raise top inequality while reducing routine-worker inequality.
- Lower learning costs decentralize problem solving and raise inequality within occupational groups.
- The empirical discussion uses U.S. wage and organization patterns as suggestive evidence for the model's distinction between information-processing technology and communication technology.

8 Conclusion

8.1 Core conclusion

The paper shows that organization is a central determinant of wage inequality. A hierarchy is not merely a production arrangement; it is a mechanism for allocating knowledge acquisition and communication across heterogeneous agents.

In equilibrium, the least able agents specialize in routine production, intermediate agents solve less common problems, and the most able agents solve the rarest problems at the top. This structure is positively sorted, stratified by ability, and supported by a convex wage function.

8.2 Economic intuition

The key economic force is the efficient use of scarce expert time. Experts should not waste time on easy questions. Lower-level agents therefore learn the routine problems and pass only exceptions upward. This makes the knowledge of high-ability agents more valuable because it can be leveraged over many subordinates.

At the same time, organization changes incentives to learn. Low-skill workers can ask instead of learning, while high-skill managers learn more because their knowledge is used repeatedly. This asymmetry is the source of the model's link between hierarchy and inequality.

8.3 Takeaway

The paper's most important lesson is that technology affects inequality through organization. Cheaper learning and cheaper communication are both forms of information technology, but they push organizations in different directions and have different effects on the wage distribution.

- Cheaper learning makes more agents knowledgeable and raises within-group inequality.
- Cheaper communication lets experts leverage their knowledge more widely and raises top inequality.

A complete theory of wage inequality therefore needs a theory of firms, hierarchies, and the internal allocation of knowledge.